

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester 3)
DSA

Practical 8

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Practical 8A

Sorting Algorithms:

```
def bubbleSort(array):
```

```
    for i in range(len(array)):
```

```
        for j in range(0, len(array) - i - 1):
```

```
            if array[j] > array[j + 1]:
```

```
                temp = array[j]
```

```
                array[j] = array[j + 1]
```

```
                array[j + 1] = temp
```

```
data = [-2, 45, 0, 11, -9]
```

```
bubbleSort(data)
```

```
print("Sorted Array in Ascending Order:")
```

```
print(data)
```

Output:

b. Insertion Sort

Code:

```
def insertionSort(array):

    for step in range(1, len(array)):

        key = array[step]
        j = step - 1

        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        array[j + 1] = key

data = [9, 5, 1, 4, 3]

insertionSort(data)

print("Sorted Array in Ascending Order:")
print(data)
```

Output:

Sorted Array in Ascending Order:
[1, 3, 4, 5, 9]

Sorted Array in Descending Order:
[9, 5, 4, 3, 1]

c. Selection Sort

Code:

```
def selectionSort(array, size):

    for step in range(size):

        min_idx = step

        for i in range(step + 1, size):

            if array[i] > array[min_idx]:
                min_idx = i

        (array[step], array[min_idx]) = (array[min_idx], array[step])

data = [-2, 45, 0, 11, -9]
size = len(data)

selectionSort(data, size)

print("Sorted Array in Descending Order:")
print(data)
print("Amit 44")
```

Output:

Sorted Array in Descending Order:
[45, 11, 0, -2, -9]

Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]